HX-240 Operating and Maintenance Manual

Controlled technical manual

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1. Scope and safety

This manual applies to the HX-240 Standard Process Pump in standard Helios configurations.

Installation, commissioning and maintenance must be performed by qualified personnel. Isolate electrical and process energy before service.

2. Operating envelope

Parameter	Limit / guidance
Flow range	15-85 m^3/h
Head	up to 62 m
Maximum system pressure	10 bar
Pump-body continuous temperature	95°C continuous fluid temperature
Suspended solids	Maximum suspended solids size 2 mm

3. Seal configuration

For continuous operation above 90°C, standard EPDM seal is not recommended.

Standard configuration: EPDM mechanical seal. Standard continuous-temperature guidance: 90°C continuous.

Available alternatives: FKM up to 120°C; PTFE up to 140°C subject to application review.

Seal life is influenced by fluid chemistry, shaft speed, pressure, temperature cycling and dry-running events.

4. Installation

• Install on a rigid, level foundation.

• Align coupling after piping loads are connected.

• Avoid unsupported pipework loads at suction and discharge nozzles.

• Provide adequate suction conditions and avoid operation outside the recommended flow region.

• Confirm direction of rotation before introducing process fluid.

5. Commissioning checklist

Check	Acceptance criterion
Process isolation removed	Confirmed safe to open
Pump primed	Casing full; no dry running
Seal flush / support	Configured if required
Coupling alignment	Within installation tolerance
Temperature and pressure	Within documented limits

6. Maintenance

Routine inspection every 4,000 operating hours; seal inspection every 2,000 hours in hot chemical service

Record operating temperature, pressure, vibration observations and seal leakage during planned inspections.

7. Troubleshooting

Symptom	Likely causes	First checks
Low flow	Incorrect rotation; restriction; air ingress	Check rotation, valves, suction line
High vibration	Misalignment; cavitation; pipe strain	Check alignment, NPSH conditions, supports
Seal leakage	Seal wear; incompatible material; dry running	Review service conditions and seal selection
Overheating	Operation outside envelope; poor cooling	Check duty point and fluid temperature